

NEERAJ KUMAR SINGH

Contact

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Profiles

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<https://nrjsingh1.github.io/>

Summary

Experienced software engineer with 9 years of strong background in building scalable distributed systems, event-driven architectures, and cloud-native applications. Currently pursuing a Masters in Computer Science from Montclair State University, New Jersey, USA.

Skills

JAVA, SPRING, POSTGRESQL, HIBERNATE, KAFKA, SPARK, WEB DEVELOPMENT, REST, MICROSERVICES
KUBERNATES, AWS, GIT, JENKINS, MAVEN, JIRA, AGILE

Experience

Here Technologies, Bengaluru , [Senior Software Engineer](#) [Sept 2021 - May 2025](#)

Spark Based Road Attribute Automation

- Migrated road attribute automation application from an event-stream-based processing system to a Spark-based batch processing job.
- Reduced processing time from over 24 hours to approximately 4 hours by leveraging Spark's MapReduce architecture and distributing large data loads across multiple worker nodes in AWS EMR clusters.
- Integrated the batch job into the Directed Acyclic Graph (DAG) execution plan, ensuring coordination with upstream and downstream processes for successful execution.

Event Stream-Based Processing

- Developed an event-driven streaming application using Java, Spring Boot, PostgreSQL, and Apache Kafka to process real-time data efficiently.
- Implemented a blocking queue mechanism to temporarily store incoming events from upstream systems and persist them in an RDBMS for intermediate storage.
- Designed and scheduled event processing workflows to handle events sequentially and publish processed data to downstream consumers.

Cisco India, Chennai , [Software Engineer II](#) [Feb 2020 - Sept 2021](#)

Cisco Cloud Calling

- Designed and implemented RESTful and SOAP-based APIs using Spring Boot and PostgreSQL, ensuring efficient and scalable backend services.
- Integrated REST clients with an Angular-based front-end, facilitating seamless communication between back-end and UI components.
- Developed a Spring-based authentication service to enhance security and user management.
- Automated UI testing using Selenium, improving application reliability and reducing manual testing efforts.

Verizon India, Chennai , [Analyst System Developer](#) [July 2019 - Feb 2020](#)

Conversation Manager Tool

- Worked on the backend of a SaaS product to design conversation flows, using Java Postgres RESTful services.

Verizon India, Chennai , [Software Engineer Developer](#) [July 2016 - July 2019](#)

Wireline Voice Portal - IVR

- Worked as Java developer in Verizon's Voice portal team, focused on enhancing DIY/ self-service and reducing call misdirects.

- Developed menu and DTMF based IVR call flows to provide self-service using VXML (Voice XML) and Nuance grammars.

Angular JS based Metrics Dashboard

- Created Voice Portal Metrics Dashboard, showing the real-time call volumes for different kinds of transaction.

Security Vulnerability Finder Application:

- Java based text scanning tool to scan the application logs to find CPNI based customers sensitive data and mask them before sending to third parties.

Verizon Store Locator App:

- Created webapp to find the nearest Verizon store from the customer location , using Php , CSS , HTML and Google map API.

Computer Science Department, CUSAT , [Internship](#)

[July 2015 - April 2016](#)

Computer Aided Process Planning, HMT Kochi

- Developed Computer Aided Process Planning python tool to read CAD diagrams of machine tools , using dxfgrabber library
- used RDBMs to store mapping of basic shapes with their planning steps.
- Generated sequence of processes required in manufacturing of machine tools
- Worked in collaboration with Hindustan Machine Tools (HMT Kochi)

College Website, School Of Engineering, Kochi

- Developed official website for college - soe.cusat.ac.in , using PHP, HTML , CSS, JQuery , MySQL, Tom-cat.
- Publish events, news and updates as updated by Admins
- Created profiles for Professors showing published papers and researches. Login and Authentication for all users.

Education

Montclair State University, New Jersey, USA ,

[2025 - 2027](#)

Master of Computer Science

Cochin University of Science & Technology, Kerala, India ,

[2012 - 2016](#)

Bachelor of Technology, Computer Science Engineering

St Michael's High School, Patna, Bihar, India ,

[2009 - 2011](#)

All India Senior School Certificate Examination (AISSCE) (12th), CBSE

Mother's International Academy, Patna, Bihar, India ,

[2009](#)

All India Secondary School Examination (10th), CBSE